

Why the Torsion Deficit Sees $H^{1/2}$, Not H^1 , and What the Capillary Obstruction Actually Rules Out

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Abstract

This note answers the question whether the Cauchy–Serrin defect D_H , together with the equation $-\Delta u = 1$, should give more than $H^{1/2}$ -smallness of a boundary perturbation. The conclusion is: boundary D_H , if available in a controlled graph regime, is indeed an H^1 -scale quantity. But the torsion deficit controls only an integrated level-set D_H , not the boundary D_H . High-frequency capillary ripples can have small amplitude and prescribed $H^{1/2}$ size while their torsion deficit is much smaller than the square of that $H^{1/2}$ size. Thus a bare $H^{1/2}$ -coercive perturbative theorem is false. This does not refute the main sharp stability theorem, whose target is $\delta_T(\Omega) \gtrsim \mathcal{A}(\Omega)^2$, since Fraenkel asymmetry is an L^1 -scale quantity rather than an $H^{1/2}$ -scale quantity.

1 Three different quantities

Let u_Ω solve

$$-\Delta u_\Omega = 1, \quad u_\Omega \in H_0^1(\Omega),$$

and write $T(\Omega) = \int_\Omega u_\Omega$. At fixed volume the ball maximizes torsion, and the Saint–Venant deficit is comparable to

$$\delta_T(\Omega) = T(B_1) - T(\Omega)$$

up to the harmless sign and factor conventions used elsewhere in the project.

For a smooth near-ball

$$\partial\Omega_\varepsilon = \{(1 + \varepsilon\varphi(\theta))\theta : \theta \in \mathbb{S}^{n-1}\},$$

with volume and barycenter normalized, the first shape derivative satisfies

$$\Delta u_1 = 0 \quad \text{in } B_1, \quad u_1|_{\partial B_1} = \frac{\varphi}{n}.$$

If $\varphi = \sum_{k \geq 0} \varphi_k$ is the spherical-harmonic decomposition, then

$$u_1(r, \theta) = \frac{1}{n} \sum_{k \geq 0} r^k \varphi_k(\theta), \quad \int_{B_1} |\nabla(nu_1)|^2 = \sum_{k \geq 1} k \|\varphi_k\|_{L^2(\mathbb{S}^{n-1})}^2.$$

This is exactly the $H^{1/2}$ scale.

On the other hand, the boundary normal derivative has first variation

$$|\nabla u_{\Omega_\varepsilon}|_{\partial\Omega_\varepsilon} = \frac{1}{n} - \frac{\varepsilon}{n} \sum_{k \geq 0} (k-1) \varphi_k + O(\varepsilon^2)$$

in the classical small C^2 graph regime. Therefore a boundary Serrin or Cauchy oscillation defect has the scale

$$D_H^\partial(\Omega_\varepsilon) \simeq \varepsilon^2 \sum_{k \geq 2} (k-1)^2 \|\varphi_k\|_{L^2(\mathbb{S}^{n-1})}^2,$$

which is an H^1 -scale quantity. The perimeter deficit has the same high frequency scale,

$$P(\Omega_\varepsilon) - P(B_1) \simeq \varepsilon^2 \sum_{k \geq 2} k^2 \|\varphi_k\|_{L^2(\mathbb{S}^{n-1})}^2.$$

Thus there is no contradiction:

$$\text{torsion deficit} \sim H^{1/2}, \quad \text{boundary } D_H \text{ or perimeter} \sim H^1.$$

The issue is that the sharp stability problem gives the former, not the latter.

2 Why the global D_H control does not give boundary H^1

The level-set identity used in Plan 2 controls an integral of the form

$$\int (D_H(t) + D_I(t)) w(t) dt \leq C \delta_T(\Omega),$$

or, in volume-radius variables, an analogous $d\mu = (-t'(\rho)) d\rho$ integral over level surfaces. This is an averaged statement over interior levels. It is not a pointwise bound for the boundary level $t = 0$.

In a coherent smooth foliation, D_H supplies the radial kinetic part of an annular Sobolev norm and D_I supplies the angular part. A trace from a bulk H^1 -type annular norm to one endpoint gives exactly an $H^{1/2}$ -boundary norm. This is the same half-derivative that appears in the Steklov spectrum of the ball.

To obtain H^1 -smallness of the boundary graph one would need substantially more information, for example a pointwise boundary D_H^∂ bound, a uniform nondegenerate graph collar, or a theorem excluding concentration of the level-set D_H in a very thin boundary layer. The equation $-\Delta u = 1$ does smooth the solution inside the domain, but without boundary geometry it does not prevent the boundary from oscillating in a region of tiny harmonic or torsional relevance.

3 A capillary counterexample to bare $H^{1/2}$ coercivity

The too-strong low-regularity perturbative theorem would say that, for radial graphs with small amplitude and small $H^{1/2}$ norm,

$$\delta_T(\Omega_\varphi) \geq c \|\varphi\|_{H^{1/2}(\mathbb{S}^{n-1})}^2$$

with no slope, perimeter, curvature, or higher-norm hypothesis. This is false already in dimension two.

Proposition 3.1 (High-frequency capillary ripples). *Let $n = 2$. For $k \geq 2$ and $0 < a \ll 1$, set*

$$R_{a,k}(\theta) = c_a(1 + a \cos(k\theta)), \quad \Omega_{a,k} = \{re^{i\theta} : 0 < r < R_{a,k}(\theta)\},$$

where c_a is chosen so that $|\Omega_{a,k}| = |B_1|$. Then the barycenter of $\Omega_{a,k}$ is zero,

$$\|R_{a,k} - 1\|_{L^\infty} \leq Ca, \quad \|R_{a,k} - 1\|_{H^{1/2}(S^1)}^2 = cka^2 + O(a^2 + a^4),$$

but

$$0 \leq T(B_1) - T(\Omega_{a,k}) \leq Ca.$$

Consequently there are smooth volume-normalized, barycentered radial graphs with $\|\varphi_j\|_{L^\infty} \rightarrow 0$, $\|\varphi_j\|_{H^{1/2}}^2 \rightarrow 0$, and

$$T(B_1) - T(\Omega_j) = o(\|\varphi_j\|_{H^{1/2}(S^1)}^2).$$

Proof. Since

$$\frac{1}{2\pi} \int_0^{2\pi} (1 + a \cos(k\theta))^2 d\theta = 1 + \frac{a^2}{2},$$

the volume normalization is achieved by

$$c_a = (1 + a^2/2)^{-1/2} = 1 + O(a^2).$$

Thus $\|R_{a,k} - 1\|_{L^\infty} \leq Ca$, and the constant Fourier mode is only $O(a^2)$. The barycenter is proportional to

$$\int_0^{2\pi} R_{a,k}(\theta)^3 e^{i\theta} d\theta.$$

For $k \geq 2$, the function $R_{a,k}^3$ contains only Fourier modes that are multiples of k , hence this first Fourier coefficient vanishes.

The $H^{1/2}(S^1)$ estimate follows from the Fourier characterization of $H^{1/2}$: the modes $\pm k$ have amplitude comparable to a , while the constant correction contributes only $O(a^4)$. Hence

$$\|R_{a,k} - 1\|_{H^{1/2}}^2 = cka^2 + O(a^2 + a^4).$$

Finally,

$$B_{1-Ca} \subset \Omega_{a,k} \subset B_{1+Ca}.$$

Torsion is monotone under inclusion, and the ball maximizes torsion at fixed volume. Therefore

$$0 \leq T(B_1) - T(\Omega_{a,k}) \leq T(B_1) - T(B_{1-Ca}).$$

In two dimensions $T(B_R) = \pi R^4/8$, so the last quantity is at most Ca .

Choose $a_j = \varepsilon_j^3$, $k_j \simeq \varepsilon_j^{-4}$, with $\varepsilon_j \rightarrow 0$. Then

$$\|R_{a_j,k_j} - 1\|_{H^{1/2}}^2 \simeq \varepsilon_j^2, \quad T(B_1) - T(\Omega_{a_j,k_j}) \leq C\varepsilon_j^3,$$

which proves the claim. $\square$

Remark 3.2. *This does not contradict the classical second variation formula. That formula is local in a stronger topology, such as small $C^{2,\gamma}$ or at least a geometry that prevents the slope from blowing up. In the example above $\|R'_{a,k}\|_{L^\infty} \sim ak$, which tends to infinity for the chosen sequence.*

4 What the equation $-\Delta u = 1$ can and cannot do

The transformation

$$h(x) = u_\Omega(x) + \frac{|x|^2}{2n}$$

is useful: h is harmonic in Ω , and the first variation at the ball is the harmonic extension of the boundary displacement. This is the reason the natural spectral norm is $H^{1/2}$.

However, harmonicity is an interior regularity statement. It does not make surface measure comparable to harmonic measure, and it does not force the solution to see every oscillation of a rough boundary. In the capillary example, the oscillations live in a thin, high-frequency boundary layer. The torsion function can be very close to the ball torsion away from that layer, and the cost of the layer is controlled by its amplitude, not by the $H^{1/2}$ size of the radial parametrization.

This is also the warning for two-dimensional conformal methods. Dirichlet energy is conformally natural, but the radial $H^{1/2}$ norm in the geometric angle θ is not automatically comparable to the conformal trace norm unless one has quantitative control of the boundary map (for instance chord-arc, quasidisk, or quasisymmetry constants). Without such control, conformal coordinates compress the capillary fingers; the bad oscillation may have very small harmonic measure.

5 Compatibility with the main asymmetry theorem

The previous proposition must not be read as a counterexample to the main theorem. In the same example,

$$|\Omega_{a,k} \Delta B_1| = \frac{1}{2} \int_0^{2\pi} |R_{a,k}(\theta)^2 - 1| d\theta = c a + O(a^2),$$

with $c > 0$ independent of k . Hence

$$\mathcal{A}(\Omega_{a,k})^2 \simeq a^2.$$

The upper bound $T(B_1) - T(\Omega_{a,k}) \leq C a$ is completely compatible with the sharp Saint–Venant conclusion

$$T(B_1) - T(\Omega_{a,k}) \geq c \mathcal{A}(\Omega_{a,k})^2 \simeq c a^2.$$

What fails is only the stronger estimate with $\|R_{a,k} - 1\|_{H^{1/2}}^2 \simeq k a^2$ on the right.

6 What extra “tame capillary” assumptions could work

The obstruction above rules out bare $H^{1/2}$ -norm coercivity, but not the main theorem and not every perturbative route. The following strengthened statements remain plausible or already available.

- (A) Classical near-spherical regime.** If $\|\varphi\|_{C^{2,\gamma}}$ is sufficiently small, BDV’s near-spherical theorem gives

$$\delta_T(\Omega_\varphi) \geq c(n) \|\varphi\|_{H^{1/2}}^2.$$

Here the coercive norm is $H^{1/2}$, but the smallness topology is much stronger.

- (B) Small $H^{1/2}$ plus a fixed high-norm bound.** If

$$\|\varphi\|_{H^{1/2}} \leq \varepsilon, \quad \|\varphi\|_{C^{m,\gamma}} \leq M(n, R)$$

with $m + \gamma > 2 + \gamma_0$, then interpolation gives

$$\|\varphi\|_{C^{2,\gamma_0}} \leq C \|\varphi\|_{H^{1/2}}^\theta M(n, R)^{1-\theta}.$$

For ε small this enters BDV. This is precisely the mechanism recorded in the graph-bootstrap notes.

- (C) Boundary D_H^∂ plus graph nondegeneracy.** If the actual boundary Serrin defect is controlled and the boundary is already a nondegenerate graph so that the expansion of $|\nabla u|$ is valid, then

$$D_H^\partial \sim \|(\Lambda - I)\varphi\|_{L^2}^2$$

does give H^1 -scale control of the non-neutral modes. The missing input in the global problem is the boundary D_H^∂ bound itself.

- (D) Uniform chord-arc, Lipschitz, NTA, or harmonic-measure control.** In dimension two, a conformal or transport method could plausibly work under a quantitative assumption preventing capillary screening, for example a uniform quasidisk/chord-arc constant together with small amplitude. Such a condition would have to imply comparability between surface measure and harmonic measure and a one-sided Dirichlet-to-Neumann form bound. The constants would depend on the tameness parameter.

- (E) Interior levels.** The equation $-\Delta u = 1$ smooths level sets away from the boundary. Integrated D_H can provide radial kinetic control through the foliation, and a trace theorem then naturally gives $H^{1/2}$ control at a selected level. Transferring that information back to $\partial\Omega$ requires a separate boundary-layer theorem excluding precisely the capillary behavior in Proposition 3.1.

7 Conclusion

The answer to the original question is therefore twofold.

First, D_H really is stronger than $H^{1/2}$ when it is a boundary Serrin defect in a controlled graph regime: spectrally it sees $(\Lambda - I)\varphi$, hence H^1 . But the sharp torsion deficit does not control that boundary quantity. It controls an integrated level-set defect, and tracing an integrated H^1 -type quantity to a boundary naturally loses a half derivative.

Second, the equation $-\Delta u = 1$ is not enough by itself to recover the missing half derivative at the rough boundary. Capillary oscillations can be screened from the torsion function. Thus the perturbative theorem with $\|\varphi\|_{H^{1/2}}^2$ on the right is false under bare $H^{1/2}$ smallness. The main theorem for this class has a different right hand side, $\mathcal{A}(\Omega)^2$, and survives this example. A Plan 3 proof that avoids Plan 1 would have to prove an L^1/L^2 -scale capillary estimate, such as a lower bound by the radial second moment $\int_{\mathbb{S}^{n-1}} (R - 1)^2$, not a full $H^{1/2}$ -coercive estimate.